# Unit III

# UNIT III

**Local Area Network: Ethernet:** Multiple Access Links and Protocols: ALOHA, CSMA, CSMA/CD, CSMA/CA. 802.3 IEEE standard, Token Bus: 802.4 IEEE standard, Token Ring: 802.5 IEEE standard, Wireless LAN Protocols. **Logical Addressing:** IPv4 Addresses, Ipv6 Addresses. INTER

# UNIT III

**Networking Devices:** Layer 1 connections- Repeater, Hubs, Layer 2 connections- Bridges, Switches, Layer 3 connections- Routers, Gateways. **Unicast Routing Protocols:** Distance Vector Routing, Link State Routing. Internet Protocols: Class A, B, C, D, E traffic, IPV4, IPV6. Address Mapping: ARP, RARP- ICMP- IGMP.

# Contents

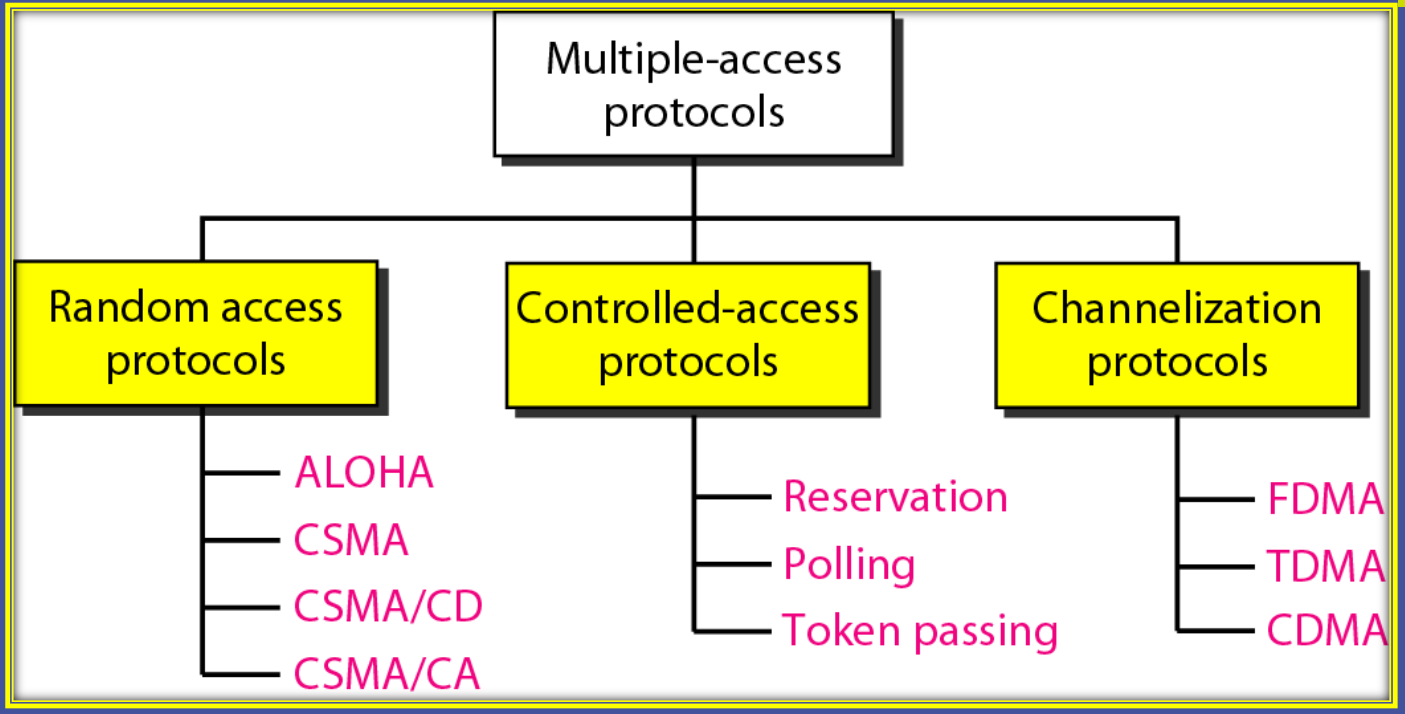


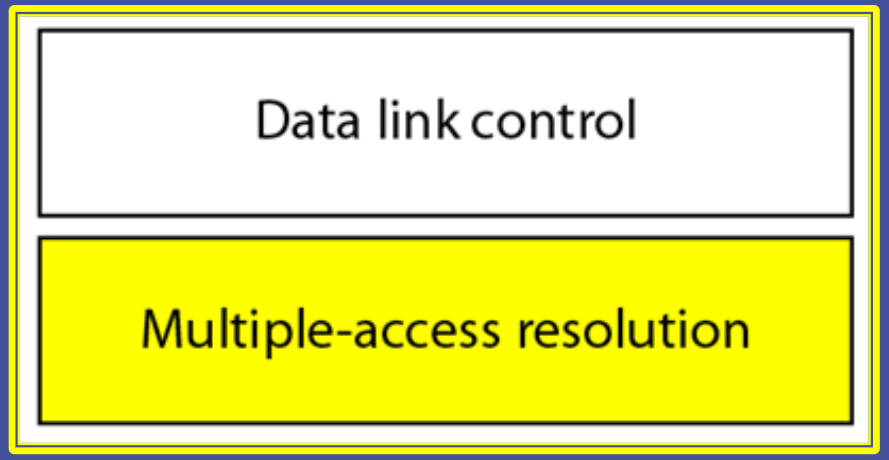


- Introduction
- Multiple Access Links and Protocols:
- ALOHA
- CSMA
- CSMA/CD
- CSMA/CA

Taxonomy of multiple-access protocols

Data link layer divided into two functionality-oriented sublayers



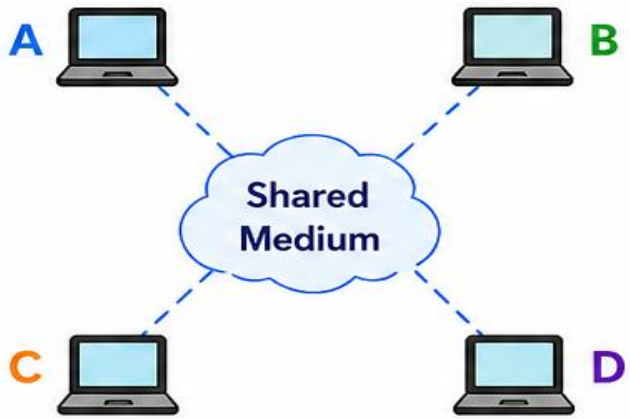
# INTRODUCTION

## Random-Access / Contention Methods

In random-access methods, **no station is superior** to another and none is assigned control over another.

### EQUAL ACCESS

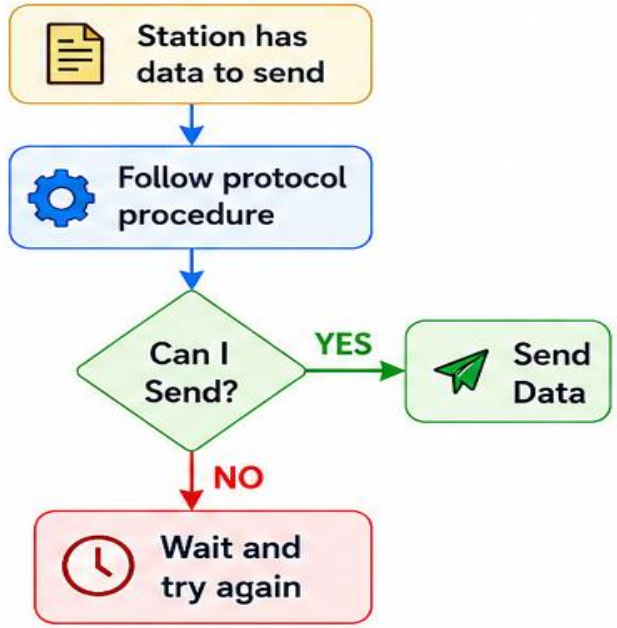
All stations have equal opportunity to send data. No station controls or manages the others.




No station is superior. All are equal.

### DECISION BY PROTOCOL

When a station has data to send, it follows a **procedure** defined by the **protocol** to decide whether or not to send.




### DEPENDS ON MEDIUM STATE

The decision to send depends on the state of the medium: **IDLE** or **BUSY.**

**IDLE (Free)**



Safe to send

**BUSY**

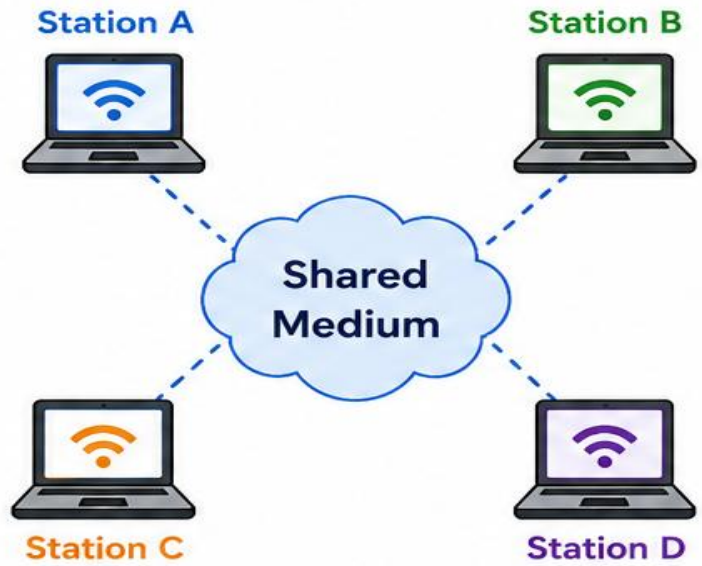


Do not send, wait

# RANDOM-ACCESS / CONTENTION METHODS

- Each station can transmit when it desires on the condition that it follows the **predefined procedure**, including testing the state of the medium.






- Transmission is **random** among the stations.




Any station can transmit at any time

- In a random-access method, each station has the **right to the medium** without being controlled by any other station.



Equal access for all

- If more than one station tries to send, **collision** occurs.




Two or more transmit at the same time

- The frames will be either **destroyed or modified.**



Original Frame

Destroyed / Modified Frame

**Key Takeaway:** In random-access methods, all stations have equal opportunity to access the medium. Collisions may happen, so stations must follow the procedure and try again if needed.



# Introduction

- In random-access or contention methods, no station is superior to another station and none is assigned control over another.
- At each instance, a station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send.
- This decision depends on the **state of the medium** (idle or busy).

# Contd...

- Each station can transmit when it desires on the condition that it follows the predefined procedure, including testing the state of the medium.
- Transmission is random among the stations

# Contd...

- In a random-access method, each station has the right to the medium without being controlled by any other station.
- If more than one station tries to send, collision occurs.
- The frames will be either destroyed or modified.

# ALOHA

## pure Aloha

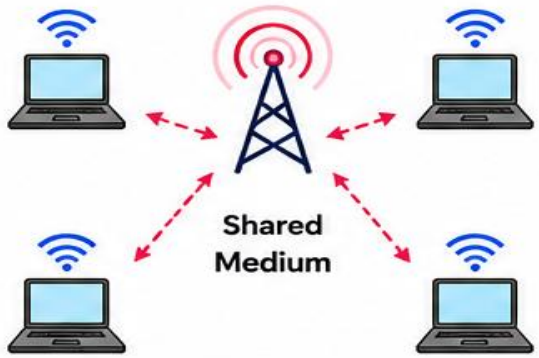
# Contd...

- ALOHA, the earliest random access method, was developed at the University of Hawaii in early 1970. It was designed for a radio (wireless) LAN - used on any shared medium.
  Collisions occurs: The medium is shared between the stations. When a station sends data, another station may attempt to do so at the same time. The data from the two stations collide and become garbled.

# ALOHA PROTOCOLS

Random access methods where all stations share the channel and transmit independently.

## ALOHA – OVERVIEW

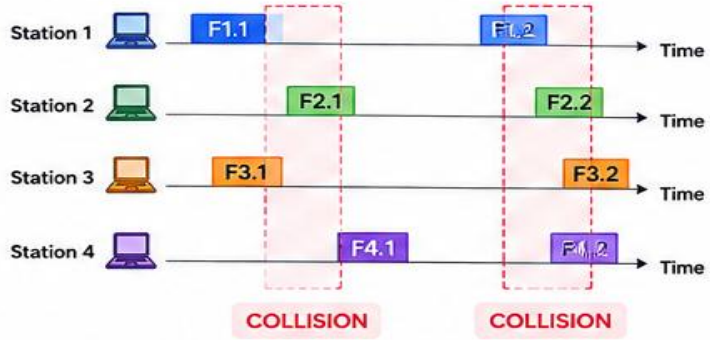



All stations have equal chance to access the shared channel. Collisions can occur.

## PURE ALOHA

- Transmit anytime
- Higher chance of collision

### HOW IT WORKS

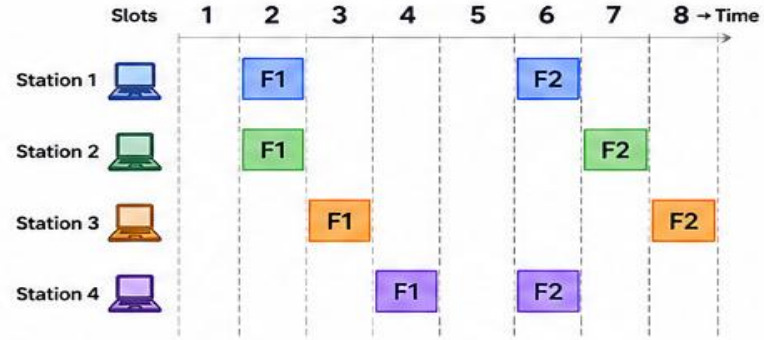



If no acknowledgment, wait a random time and retransmit.

## SLOTTED ALOHA

- Time divided into equal slots
- Transmit only at start of a slot
- Lower chance of collision

### HOW IT WORKS




Collisions only if two or more stations transmit in the same slot.

| Feature | PURE ALOHA | SLOTTED ALOHA |
|---|---|---|
| Transmission Time | Any time | At beginning of slots |
| Synchronization | Not required | Required |
| Collision Probability | Very High | Lower |
| Max Throughput (Ideal) | 18% | ~37% |

### Comparison



# Pure ALOHA

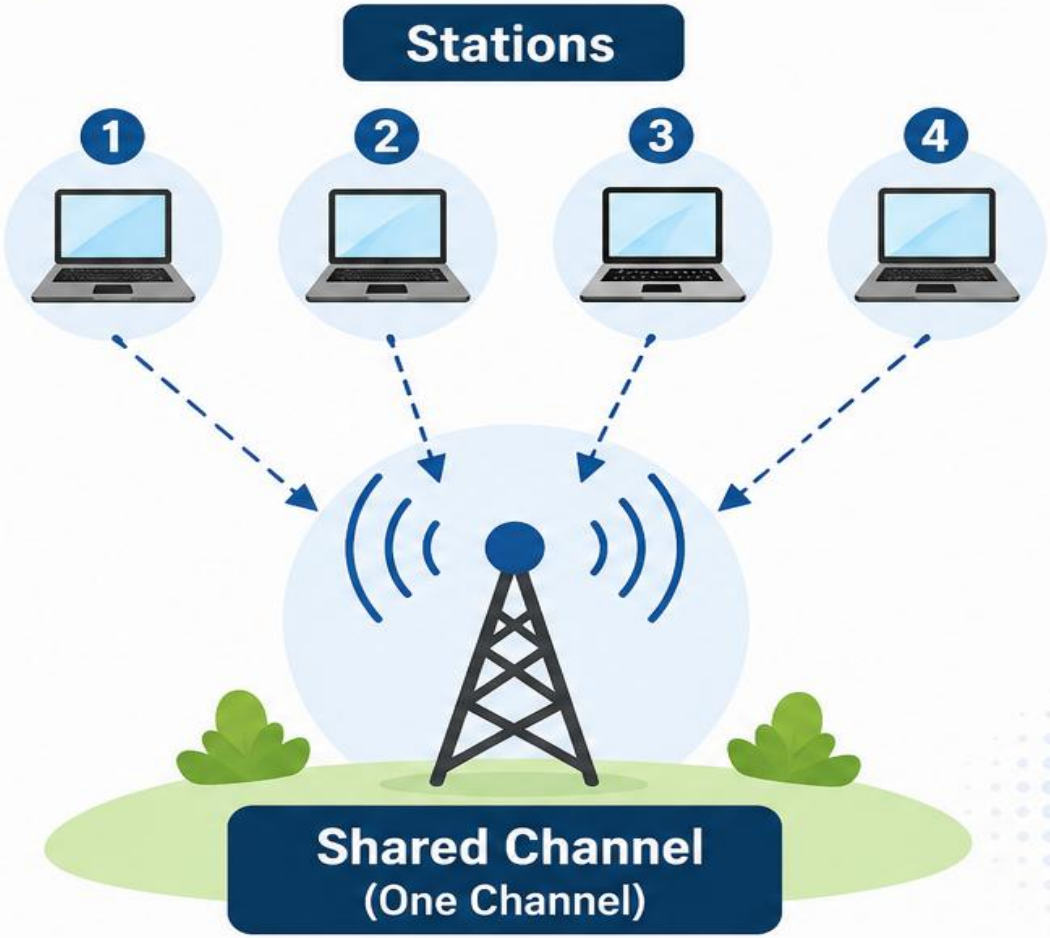
- Pure ALOHA: The original ALOHA protocol is called pure ALOHA.
- The idea is that each station sends a frame whenever it has a frame to send (multiple access).
- Since there is only one channel to share, there is the possibility of collision between frames from different stations.

# 1. OVERVIEW – PURE ALOHA

**Pure ALOHA:** The original ALOHA protocol is called pure ALOHA.

The idea is that each station sends a **frame** whenever it has **a frame to send** (multiple access).

Since there is only one channel to share, there is the possibility of **collision between frames from different stations.**



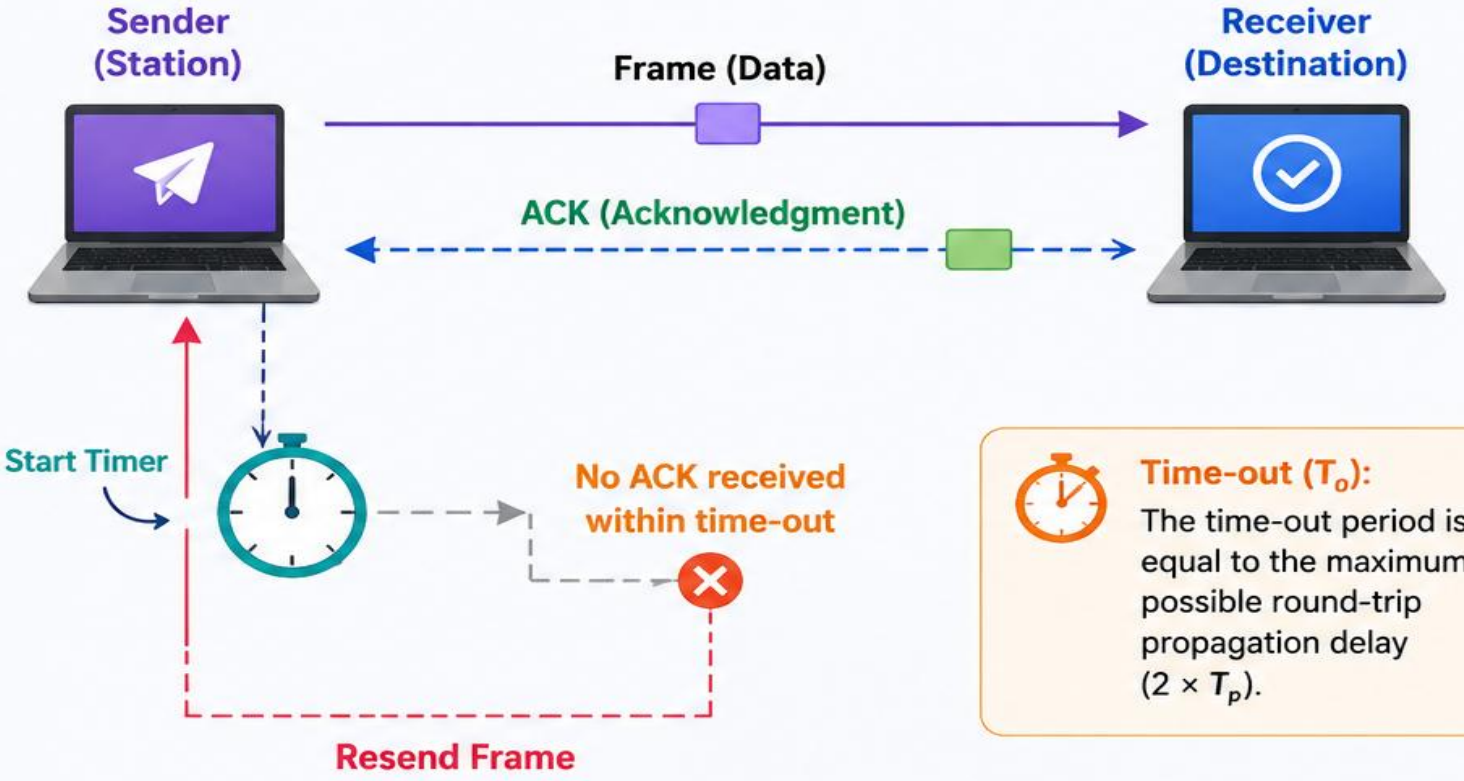

Multiple stations share the same channel. When two or more stations transmit at the same time, their frames **collide** and become **garbled (lost)**.

# ACKNOWLEDGMENT & TIME-OUT

Pure ALOHA relies on **acknowledgments** from the receiver.

When a station sends a frame, it expects the receiver to send an **acknowledgment**.

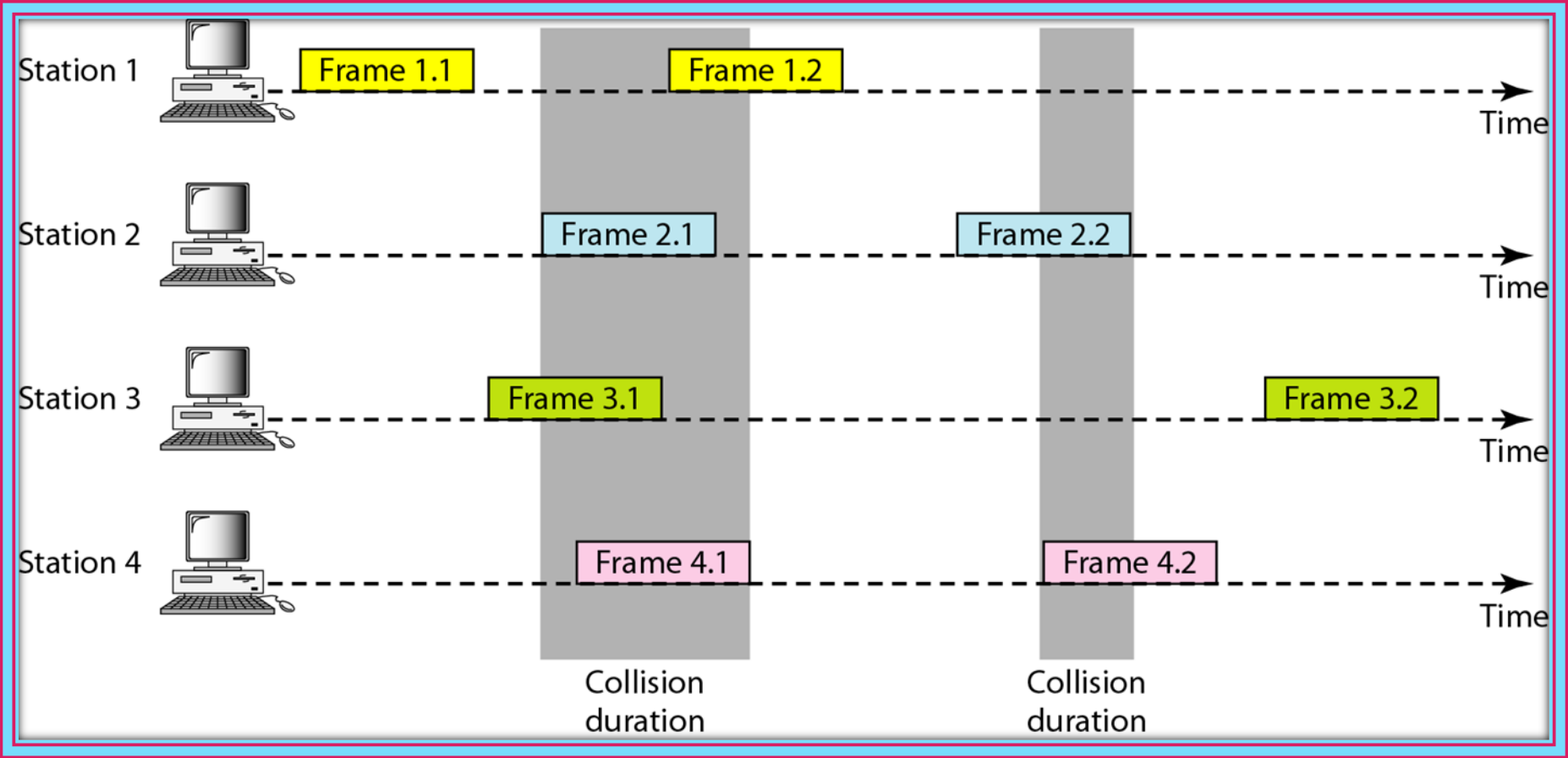
If the acknowledgment does not arrive after a **time-out period**, the station assumes that the frame (or the acknowledgment) has been destroyed and **resends the frame**.




**Time-out ($T_o$):**
The time-out period is equal to the maximum possible round-trip propagation delay ($2 \times T_p$).

**Key Idea:** Acknowledgments confirm successful delivery. If an ACK is not received within the time-out period, the frame is assumed lost and is retransmitted.

# Frames in a pure ALOHA network



# Contd…

- There are four stations (unrealistic assumption) that contend with one another for access to the shared channel.
- The figure shows that each station sends two frames; there are a total of eight frames on the shared medium. Some of these frames collide because multiple frames are in contention for the shared channel.
- Figure shows that only two frames survive: one frame from station 1 and one frame from station 3.

# Contd…

- We need to mention that even if one bit of a frame coexists on the channel with one bit from another frame, there is a collision and both will be destroyed.
- The frames that have been destroyed during transmission is resend.

# CONTD...

- The pure ALOHA protocol relies on acknowledgments from the receiver.
- When a station sends a frame, it expects the receiver to send an acknowledgment.
- If the acknowledgment does not arrive after a time-out period, the station assumes that the frame (or the acknowledgment) has been destroyed and resends the frame.

# CONTD...

- A collision involves two or more stations.
- If all these stations try to resend their frames after the time-out, the frames will collide again.
- Pure ALOHA dictates that when the time-out period passes, each station waits a random amount of time before resending its frame.
- The randomness will help avoid more collisions.
- We call this time the backoff time TB.

# CONTD...

- Pure ALOHA has a second method to prevent congesting the channel with retransmitted frames.
- After a maximum number of retransmission attempts $K_{max}$, a station must give up and try later.
- The time-out period is equal to the maximum possible round-trip propagation delay, which is twice the amount of time required to send a frame between the two most widely separated stations ($2 \times Tp$).

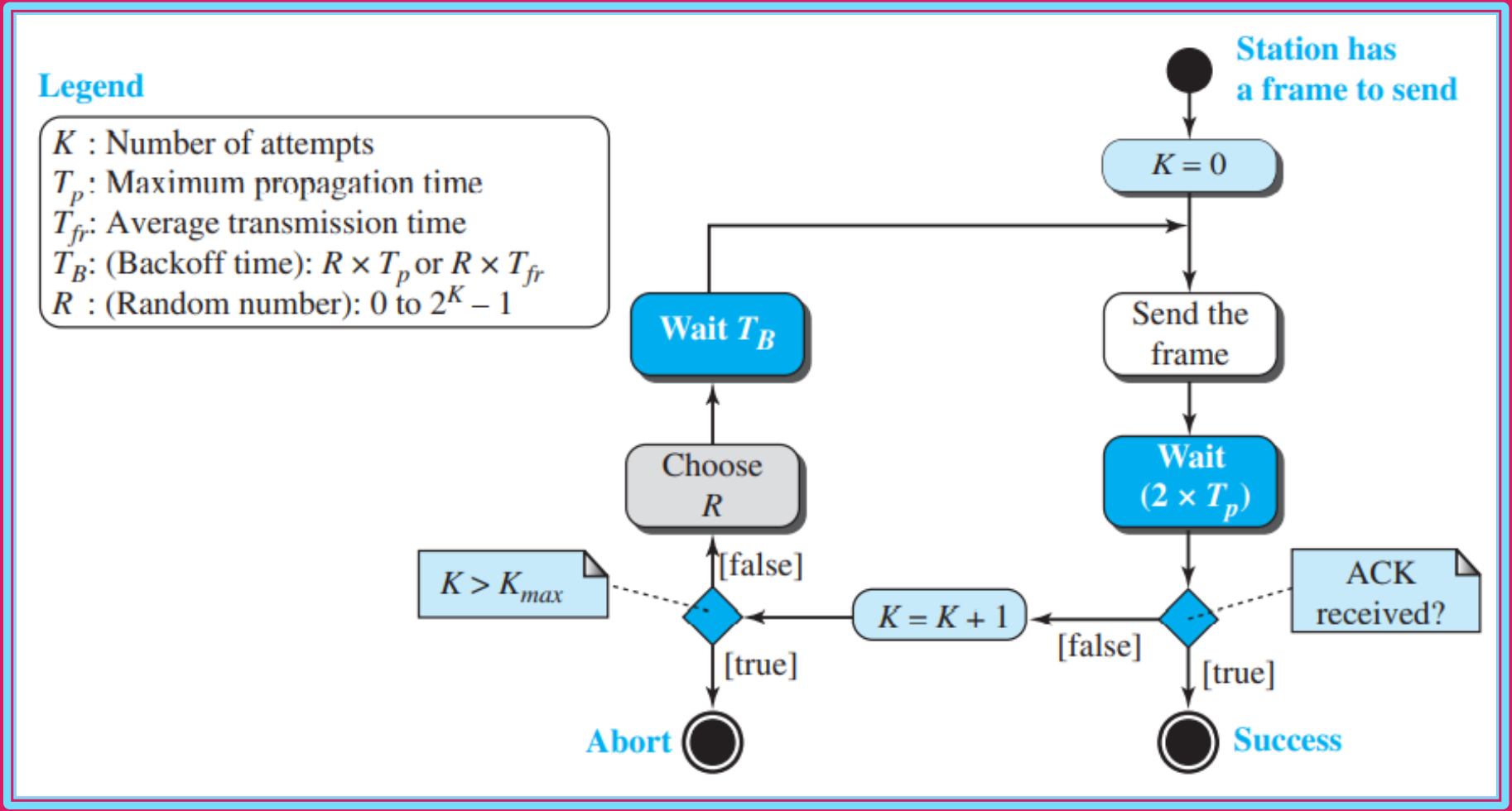
# CONTD…

- The backoff time $T_B$ is a random value that normally depends on K (the number of attempted unsuccessful transmissions).
- The formula for $T_B$ depends on the implementation.
- One common formula is the binary exponential backoff.

# CONTD...

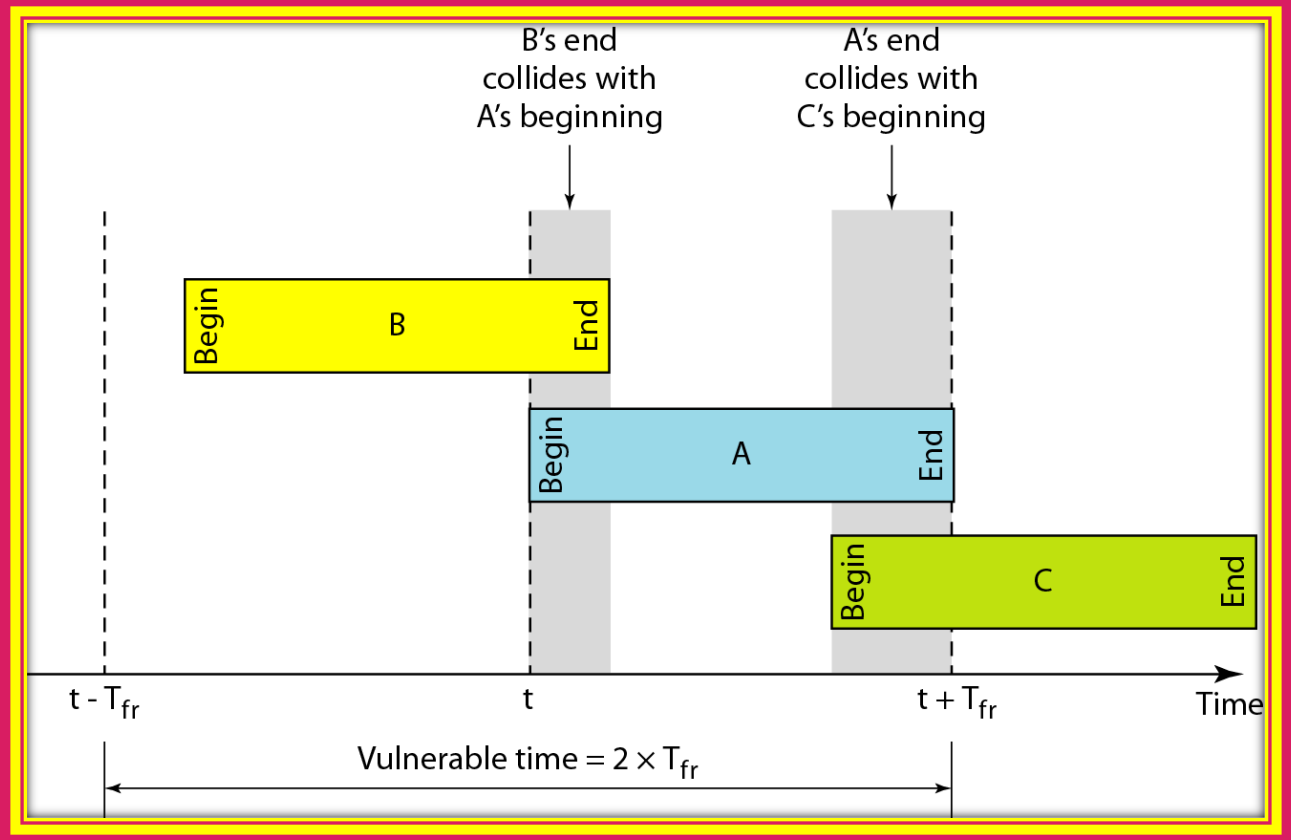
- In this method, for each retransmission, a multiplier R = 0 to $2^K - 1$ is randomly chosen and multiplied by Tp (maximum propagation time) or $T_{fr}$ (the average time required to send out a frame) to find $T_B$.
- The range of the random numbers increases after each collision. The value of $K_{max}$ is usually chosen as 15.

# Procedure for pure ALOHA protocol



# Vulnerable time for pure ALOHA protocol

vulnerable time, the length of time in which there is a possibility of collision




Pure ALOHA vulnerable time = $2 \times T_{fr}$

# CONTD...

- Station B starts to send a frame at time t. Now imagine station A has started to send its frame after $t - T_{fr}$.
- This leads to a collision between the frames from station B and station A. On the other hand, suppose that station C starts to send a frame before time $t + T_{fr}$.
- There is also a collision between frames from station B and station C.

# Problem

A Pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces

a. 1000 frames per second b. 500 frames per second c. 250 frames per second.

## PROBLEM

A Pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces

(a) 1000 frames per second (b) 500 frames per second (c) 250 frames per second



## 1. GIVEN

- Frame size (L) = 200 bits
- Channel rate (R) = 200 kbps
- Pure ALOHA
- Offered load (G) = $\frac{N \times L}{R}$ (Erlang)

N = frames generated per second
L = frame size in bits
R = channel rate in bps

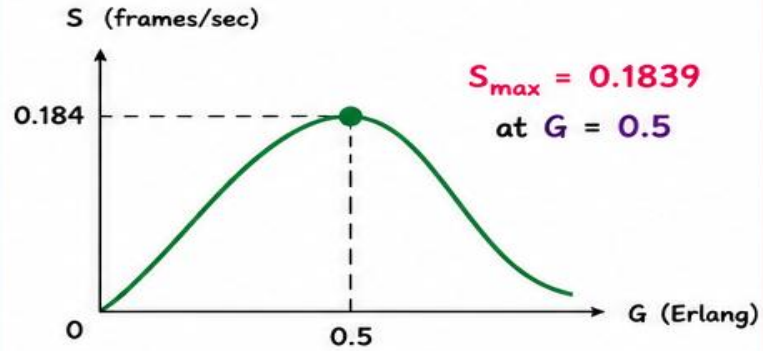
## 2. PURE ALOHA THROUGHPUT FORMULA

Throughput (S) = $G\, e^{-2G}$

(in frames per second)

Maximum throughput occurs at G = 0.5 with $S_{max}$ = 0.1839 frames/sec (18.39%)

## THROUGHPUT vs LOAD




## 3. CALCULATIONS

**a** N = 1000 frames/sec

$$G = \frac{1000 \times 200}{200{,}000} = 1 \text{ Erlang}$$

$$S = G\, e^{-2G} = 1 \times e^{-2(1)} = e^{-2}$$

S = 0.1353 frames/sec (13.53%)

**b** N = 500 frames/sec

$$G = \frac{500 \times 200}{200{,}000} = 0.5 \text{ Erlang}$$

$$S = G\, e^{-2G} = 0.5 \times e^{-2(0.5)} = 0.5 \times e^{-1}$$

S = 0.1839 frames/sec (18.39%)

**c** N = 250 frames/sec

$$G = \frac{250 \times 200}{200{,}000} = 0.25 \text{ Erlang}$$

$$S = G\, e^{-2G} = 0.25 \times e^{-2(0.25)} = 0.25 \times e^{-0.5}$$

S = 0.1516 frames/sec (15.16%)

**THROUGHPUTS** (frames/sec):

(a) 0.1353 (13.53%) | (b) 0.1839 (18.39%) | (c) 0.1516 (15.16%)

MAXIMUM THROUGHPUT occurs at (b) 500 frames/sec with $S_{max}$ = 0.1839 frames/sec

Solution (a)

The frame transmission time is 200/200 kbps or 1 ms.

a. If the system creates 1000 frames per second, this is 1 frame per millisecond. The load is 1.
b. In this case

$$S = G \times e^{-2G} \text{ or } S = 0.135 \text{ (13.5 percent).}$$

c. This means that the throughput is 1000 × 0.135 = 135 frames.
d. Only 135 frames out of 1000 will probably survive.

Solution (b)

- If the system creates 500 frames per second, this is (1/2) frame per millisecond. The load is (1/2).
- In this case $S = G \times e^{-2G}$ or $S = 0.184$ (18.4 percent).
- This means that the throughput is

$$500 \times 0.184 = 92$$

- Only 92 frames out of 500 will probably survive.

Solution (c)

- Only 38 frames out of 250 will probably survive.



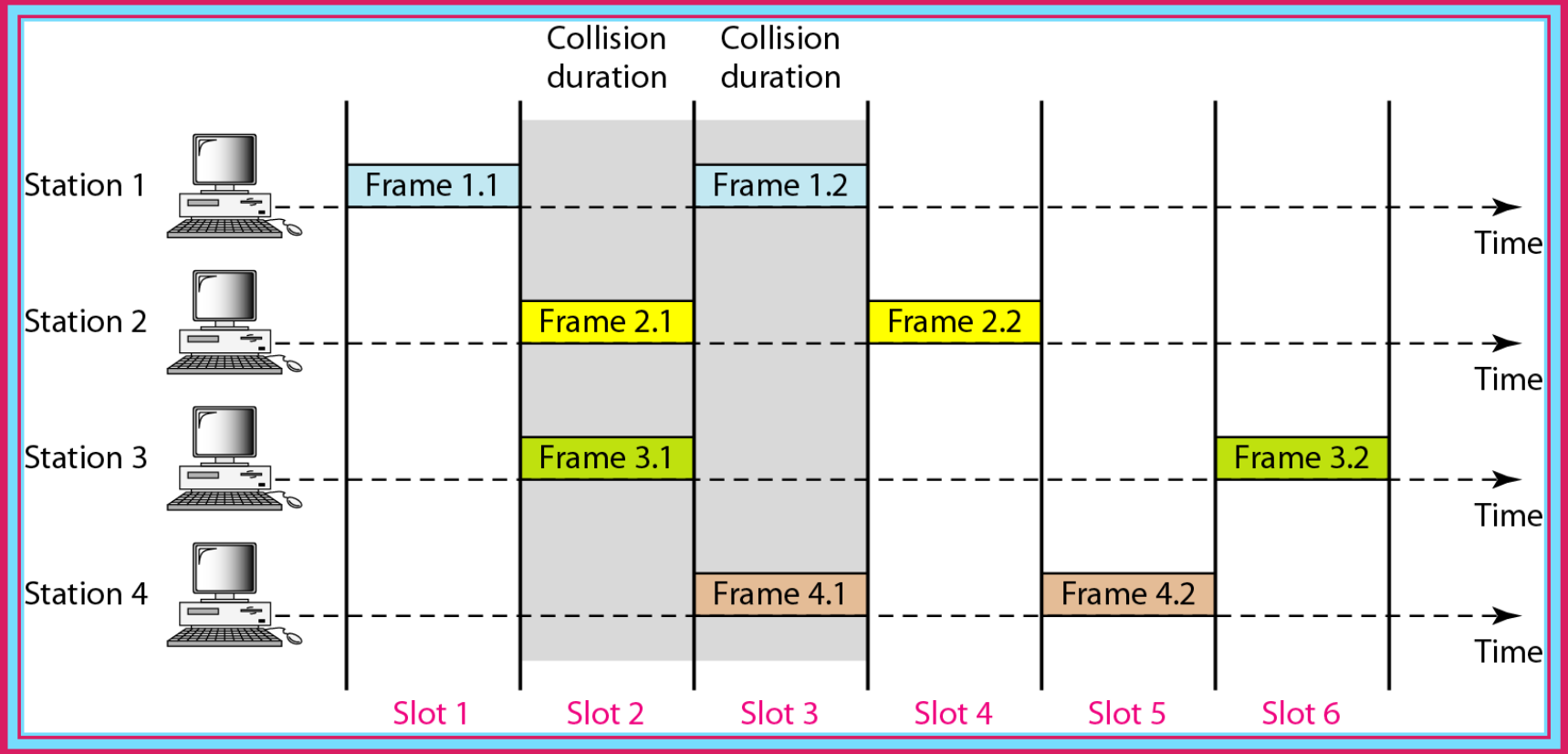
# ALOHA

## Slotted Aloha

# Slotted Aloha

- Slotted ALOHA was invented to improve the efficiency of pure ALOHA.
- In slotted ALOHA we divide the time into slots of $T_{fr}$ seconds and force the station to send only at the beginning of the time slot.
- Because a station is allowed to send only at the beginning of the synchronized time slot, if a station misses this moment, it must wait until the beginning of the next time slot.

# Frames in a slotted ALOHA network



*Note*

**The throughput for slotted ALOHA is**

$$S = G \times e^{-G}$$

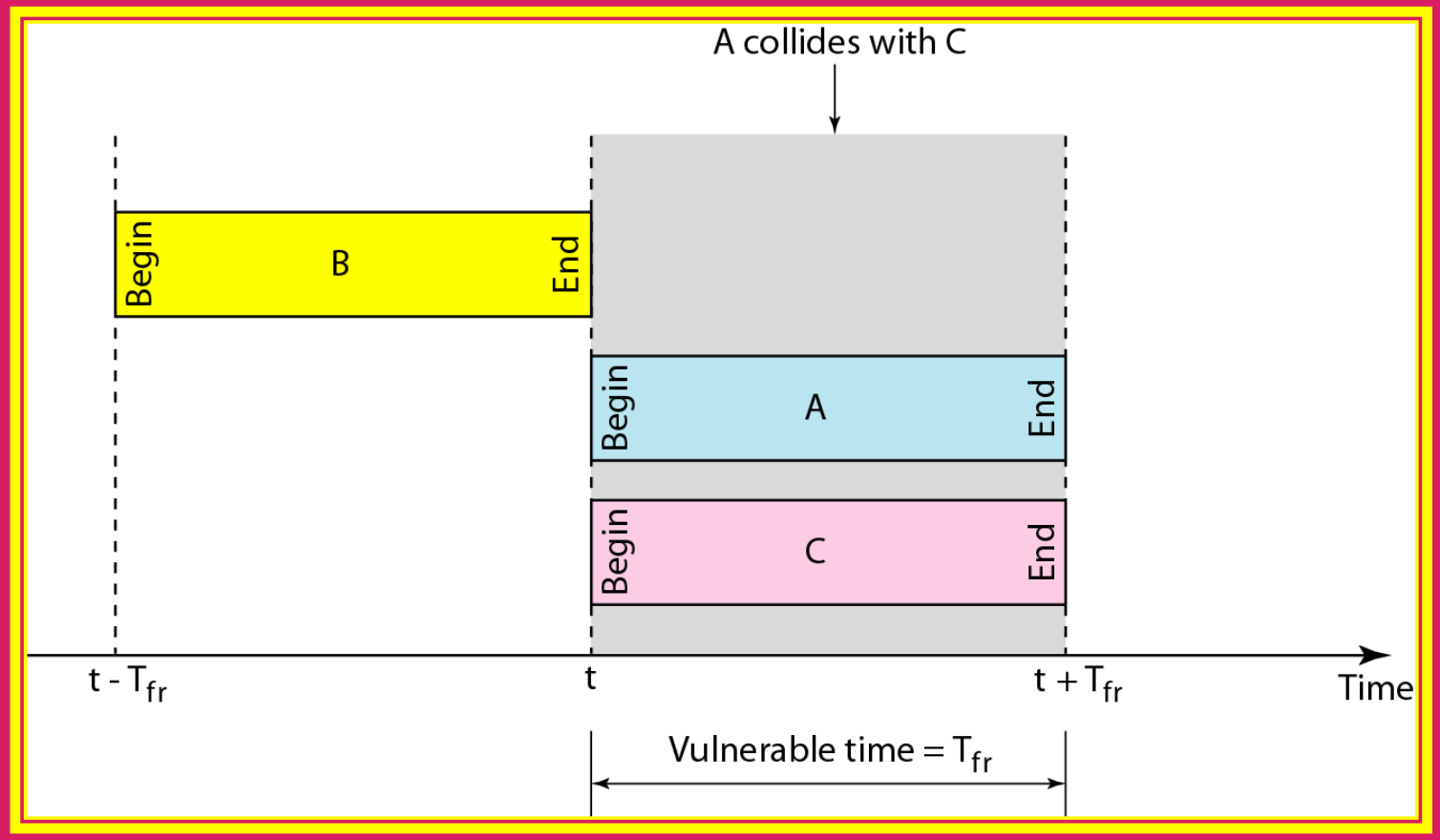
**The maximum throughput**

$S_{max} = 0.368$ **when G = 1.**

# CONTD...

- This means that the station which started at the beginning of this slot has already finished sending its frame.
- There is still the possibility of collision if two stations try to send at the beginning of the same time slot.
- The vulnerable time is now reduced to one-half, equal to $T_{fr}$.

# Vulnerable time for slotted ALOHA protocol




Slotted ALOHA vulnerable time = $T_{fr}$

# Problem

A slotted ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces

a. 1000 frames per second b. 500 frames per second c. 250 frames per second.

## PROBLEM

A slotted ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces

(a) 1000 frames per second (b) 500 frames per second (c) 250 frames per second



## KEY FACTS

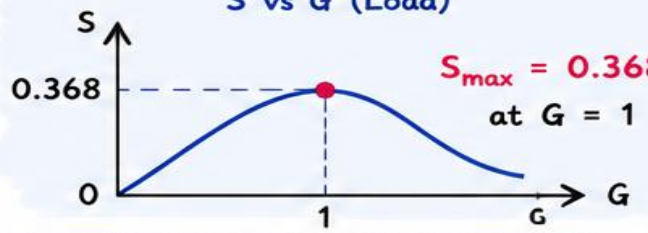
- Frame size (L) = 200 bits
- Channel rate (R) = 200 kbps
- Slotted ALOHA

## SLOTTED ALOHA THROUGHPUT FORMULA

$$S = G\, e^{-G}$$

(in frames per second)

S vs G (Load)




$S_{max} = 0.368$ at $G = 1$

Where $G = \frac{N \times L}{R}$ (Erlang) , N = frames generated per second L = frame size in bits , R = channel rate in bps

### (a) N = 1000 frames/sec

$$G = \frac{1000 \times 200}{200{,}000} = 1 \text{ Erlang}$$

$S = G\, e^{-G} = 1 \times e^{-1} = 0.368$

Throughput = 1000 × 0.368
= 368 frames/sec

Only 368 frames out of 1000 will probably survive.

### (b) N = 500 frames/sec

$$G = \frac{500 \times 200}{200{,}000} = 0.5 \text{ Erlang}$$

$S = G\, e^{-G} = 0.5 \times e^{-0.5} = 0.303$

Throughput = 500 × 0.303
= 151 frames/sec

Only 151 frames out of 500 will probably survive.

### (c) N = 250 frames/sec

$$G = \frac{250 \times 200}{200{,}000} = 0.25 \text{ Erlang}$$

$S = G\, e^{-G} = 0.25 \times e^{-0.25} = 0.779$

Throughput = 250 × 0.779
= 194.8 frames/sec

Only 49 frames out of 250 will probably survive.

## SUMMARY (THROUGHPUTS)

(a) 368 frames/sec | (b) 151 frames/sec | (c) 194.8 frames/sec

## MAXIMUM THROUGHPUT

occurs at G = 1 with
$S_{max} = 0.368$ frames/sec

## Solution (a)

The frame transmission time is 200/200 kbps or 1 ms.

a. If the system creates 1000 frames per second, this is 1 frame per millisecond. The load is 1.
b. In this case

$$S = G \times e^{-G} \text{ or } S = 0.368 \text{ (36.8 percent).}$$

c. This means that the throughput is 1000 × 0.368 = 368 frames.
d. Only 386 frames out of 1000 will probably survive.

Solution (b)

- If the system creates 500 frames per second, this is (1/2) frame per millisecond. The load is (1/2).
- In this case $S = G \times e^{-G}$ or $S = 0.303$ (30.3 percent).
- This means that the throughput is

$$500 \times 0.303 = 151$$

- Only 151 frames out of 500 will probably survive.

Solution (c)

- Only 49 frames out of 250 will probably survive.

# CSMA



# CSMA

## Carrier Sense Multiple Access

CSMA lets stations listen before they talk on the shared channel.



Principle: "Sense before transmit"

## HOW IT WORKS

**1 Listen**



Station listens to the channel.

**2 If Idle**



If channel is idle, station sends its data.

**3 If Busy**



If channel is busy, station waits and listens again.



## WHY COLLISIONS CAN STILL HAPPEN?

Due to propagation delay, two stations may sense the channel idle and start transmitting at the same time.




## KEY TAKEAWAY



CSMA **reduces** the chance of collision.



But it **cannot** eliminate collisions.



**Like driving:**
Look left and right before crossing.



Road Clear ✓



Go!

Traffic Coming ✗

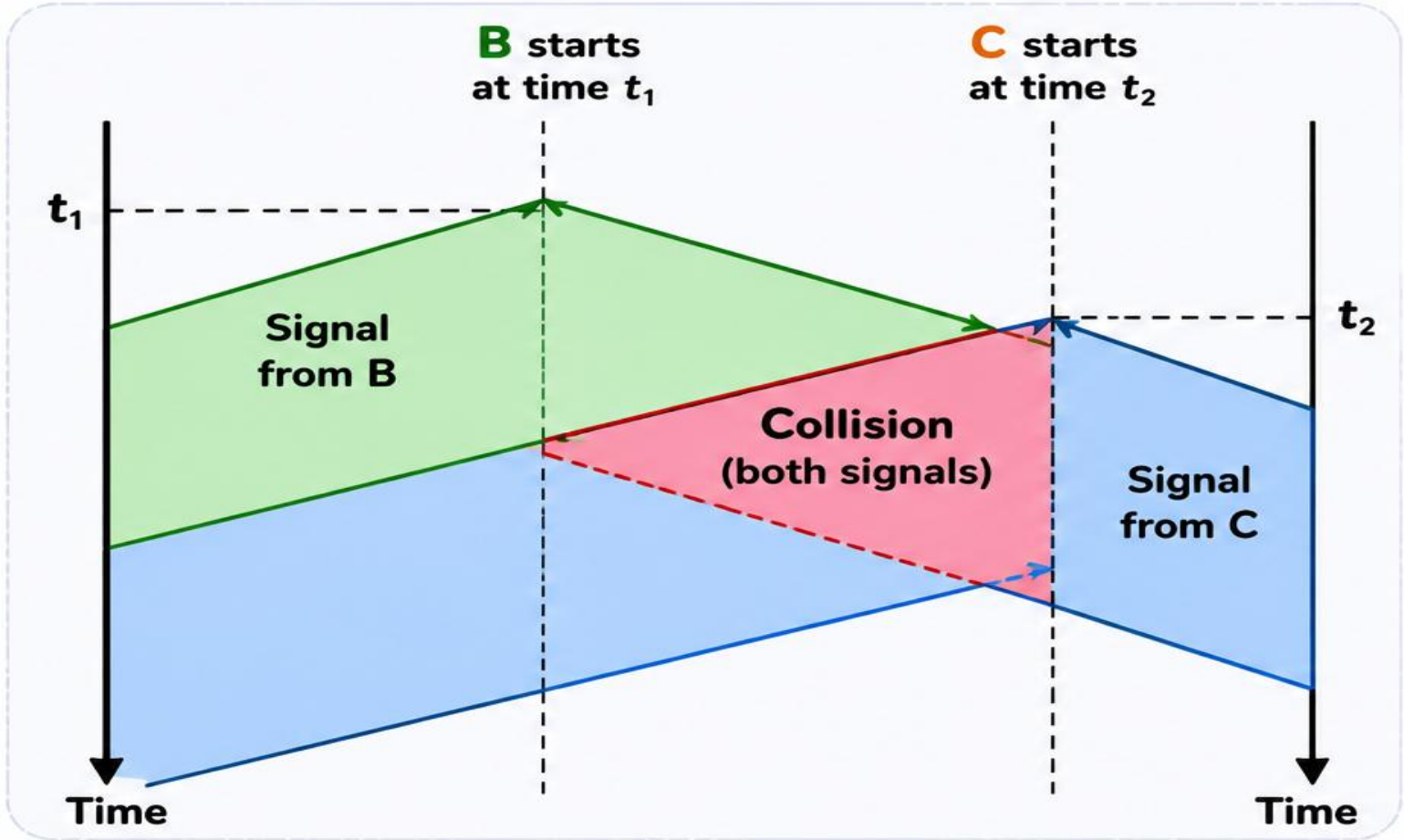


Wait!



# Space/Time Model of a Collision in CSMA

A — B — C — D




1. **B** — At time $t_1$, Station **B** senses the medium idle and sends.

2. **C** — At time $t_2$ ($t_2 > t_1$), Station **C** senses the medium idle (first bits from B not yet received) and sends.

3. The signals meet in between → **Collision**!

4. Both frames are destroyed.

**Key Idea:** Due to propagation delay, two stations may think the channel is idle and transmit at the same time → **Collision**.

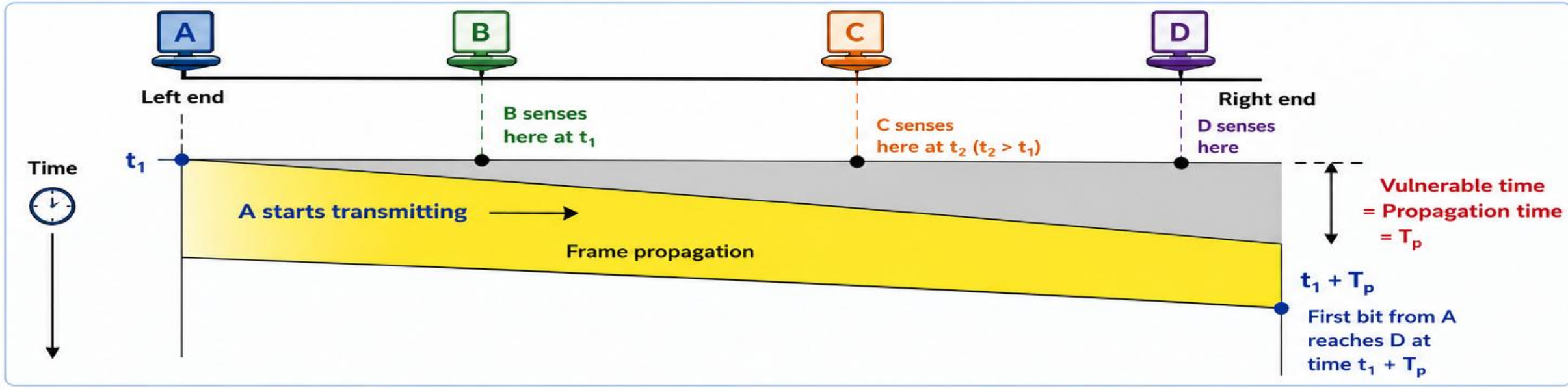
- Signal from B
- Signal from C
- Collision region

# VULNERABLE TIME IN CSMA

In CSMA, a station must listen before it talks.
But due to propagation delay, another station may start talking before it hears the first bit.

This risky period is called **VULNERABLE TIME.**

Vulnerable time = Propagation time = $T_p$




**1 A starts at time $t_1$**

A senses the channel idle and starts sending.




The signal takes time $T_p$ to reach the other end.

**2 C starts at time $t_2$ ($t_2 > t_1$)**

C also senses the channel idle (first bit from A has not yet reached C) and starts sending.




Transmission during this time → COLLISION!

**3 Collision occurs**

The two signals overlap in the medium and collide.




Both frames are destroyed.

**4 After $t_1 + T_p$**

By the time the first bit from A reaches D, all stations already know the channel is busy and refrain from sending.




Time = $T_p$

**KEY TAKEAWAY:** Vulnerable time in CSMA = Propagation time ($T_p$).
Any transmission during this time can cause a collision.

**Why it happens?**

Propagation delay – time needed for the first bit to travel from one end of the medium to the other.

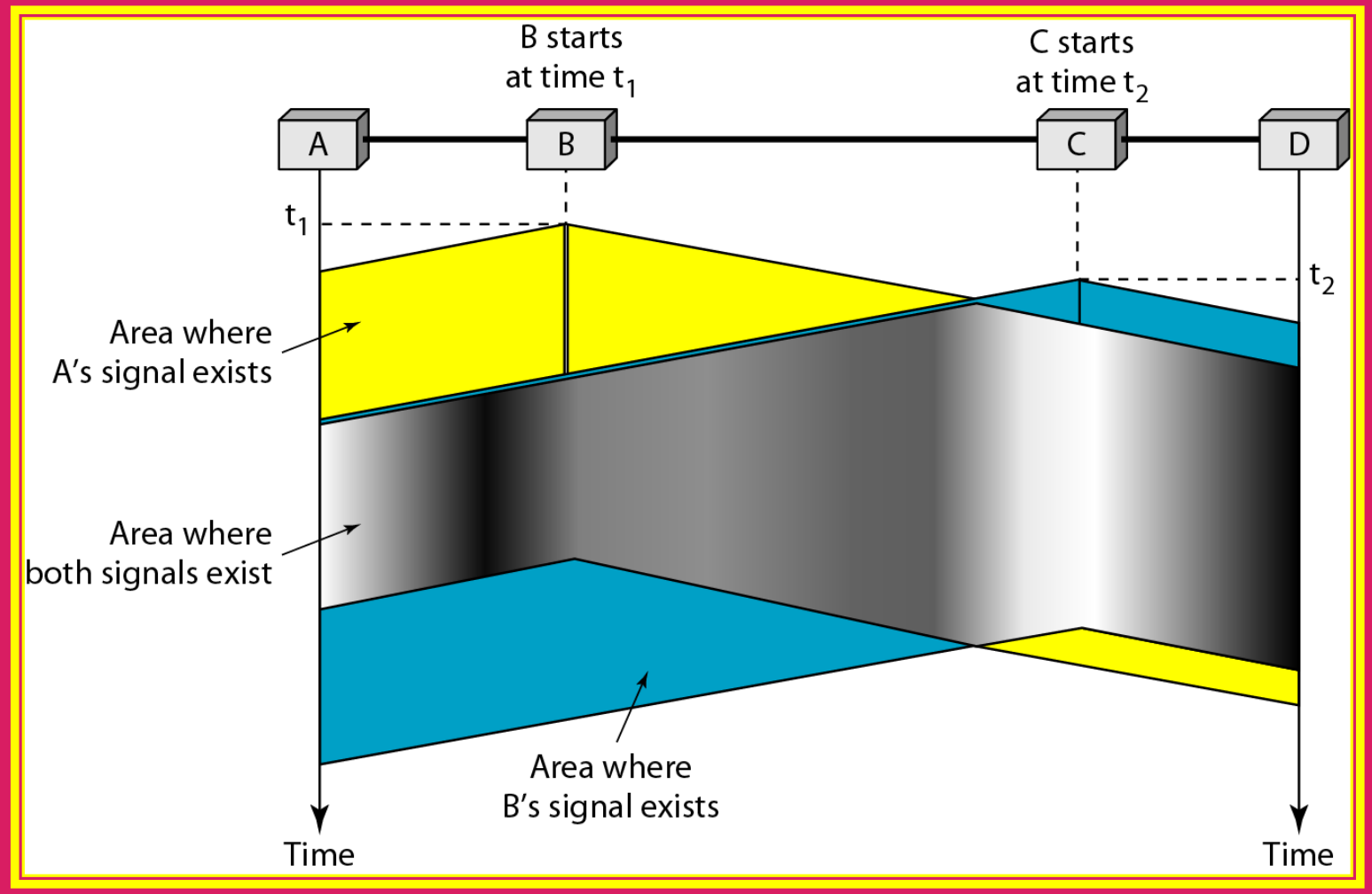
# CSMA

- Carrier sense multiple access (CSMA) requires that each station first listen to the medium (or check the state of the medium) before sending.
- In other words, CSMA is based on the principle "sense before transmit" or "listen before talk."
- CSMA can reduce the possibility of collision, but it cannot eliminate it.

# CONTD…

- Stations are connected to a shared channel (usually a dedicated medium).
- The possibility of collision still exists because of propagation delay; when a station sends a frame, it still takes time (although very short) for the first bit to reach every station and for every station to sense it.
- A station may sense the medium and find it idle, only because the first bit sent by another station has not yet been received.

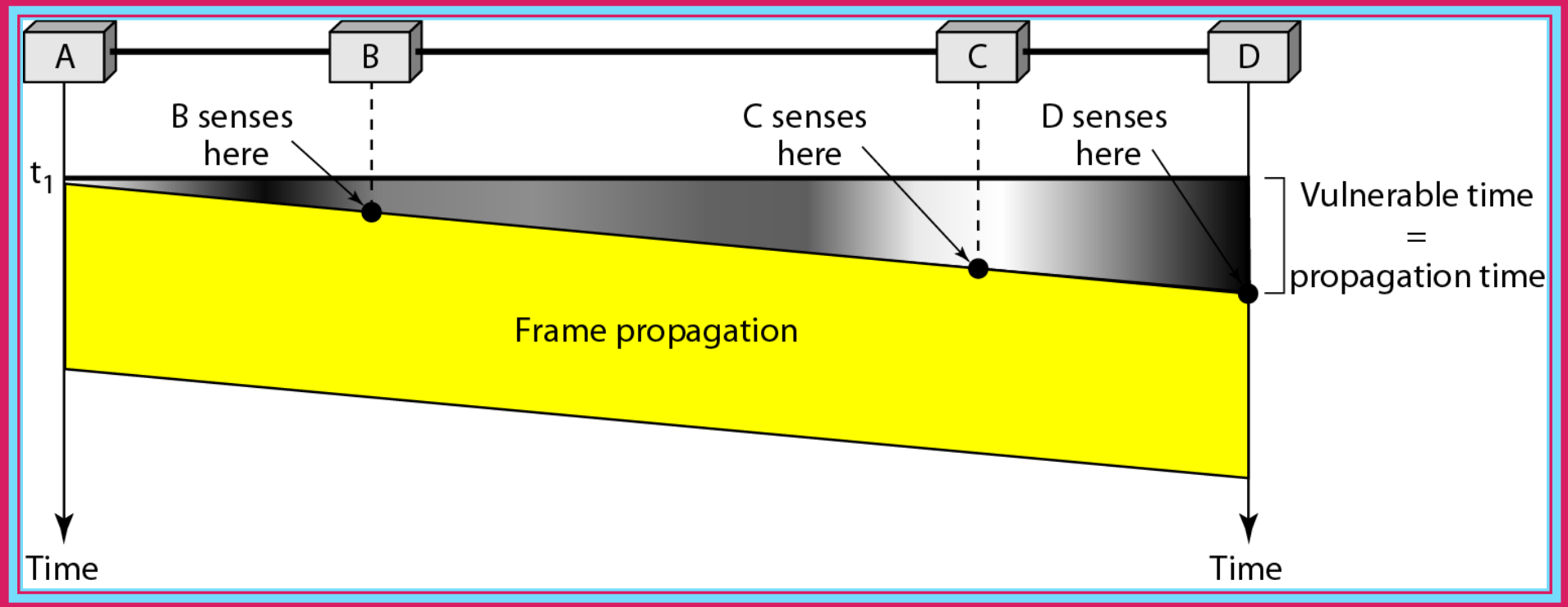
# Space/Time model of a collision in CSMA



# CONTD...

- At time $t_1$, station B senses the medium and finds it idle, so it sends a frame.
- At time $t_2$ ($t_2 > t_1$), station C senses the medium and finds it idle because, at this time, the first bits from station B have not reached station C.
- Station C also sends a frame. The two signals collide and both frames are destroyed.

# Vulnerable time in CSMA




vulnerable time in CSMA is propagation time = $T_p$

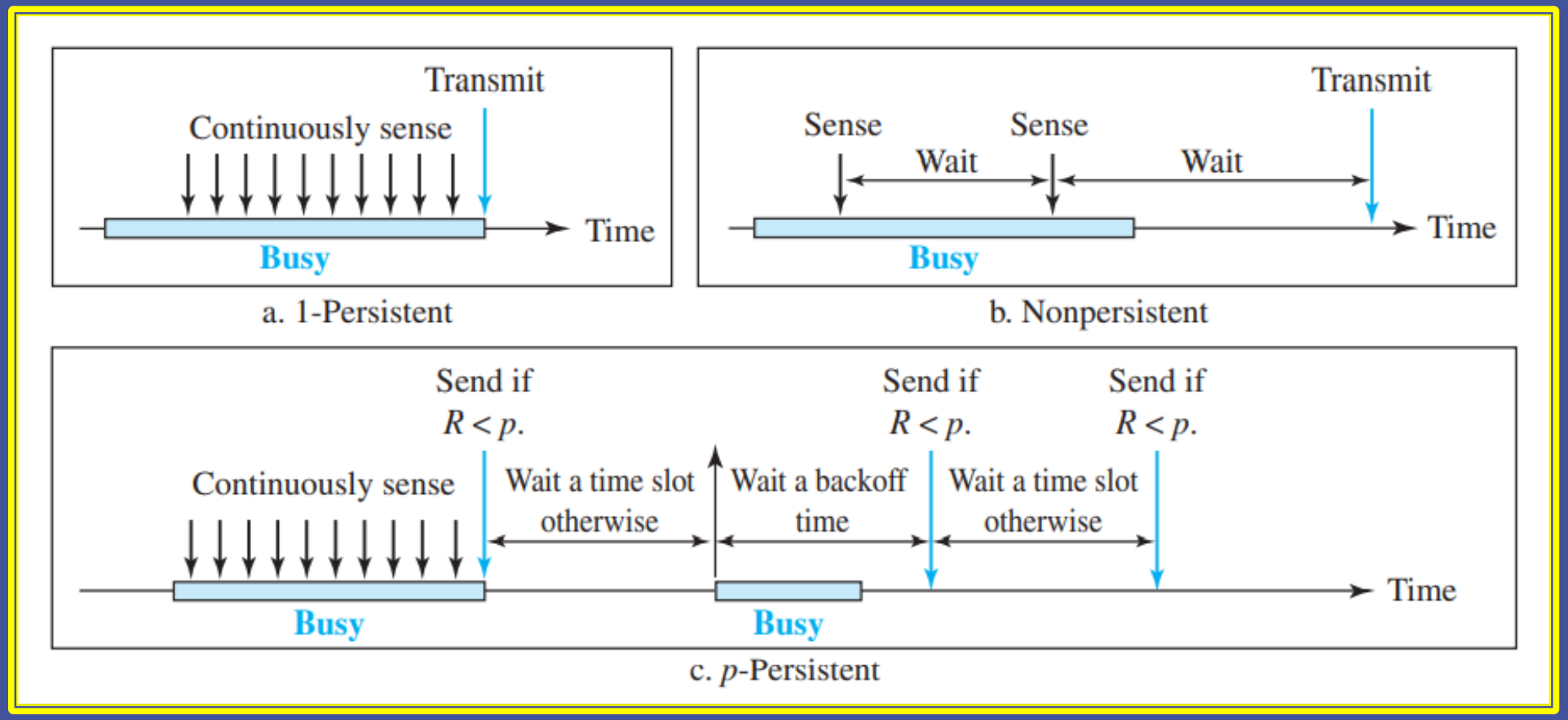
- This is the time needed for a signal to propagate from one end of the medium to the other.
- When a station sends a frame and any other station tries to send a frame during this time, a collision will result.
- But if the first bit of the frame reaches the end of the medium, every station will already have heard the bit and will refrain from sending.
- The leftmost station, A, sends a frame at time $t_1$, which reaches the rightmost station, D, at time $t_1 + T_p$. The gray area shows the vulnerable area in time and space.

# Persistence Methods

- 1-persistent method
- nonpersistent method
- p-persistent method

- The 1-persistent method is simple and straightforward.
- After the station finds the line idle, it sends its frame immediately (with probability 1).
- This method has the highest chance of collision because two or more stations may find the line idle and send their frames immediately.

# Behavior of three persistence methods



# CONTD...

- In the nonpersistent method, a station that has a frame to send senses the line.
- If the line is idle, it sends immediately. If the line is not idle, it waits a random amount of time and then senses the line again.
- Reduces the chance of collision because it is unlikely that two or more stations will wait the same amount of time and retry to send simultaneously.
- This method reduces the efficiency of the network because the medium remains idle when there may be stations with frames to send.

# CONTD…

- The p-persistent method is used if the channel has time slots with a slot duration equal to or greater than the maximum propagation time.
- The p-persistent approach combines the advantages of the other two strategies.
- It reduces the chance of collision and improves efficiency.

# CONTD...

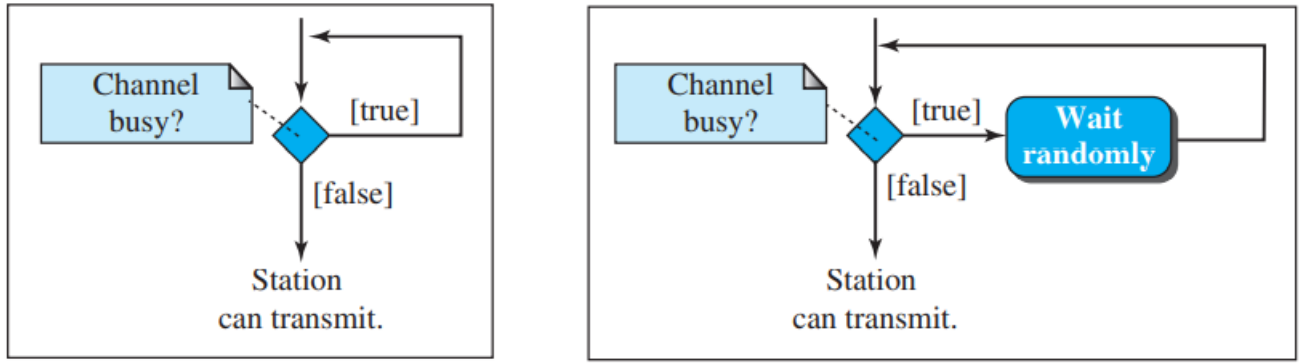
- In this method, after the station finds the line idle it follows these

Step: 1. With probability p, the station sends its frame.

Step: 2 With probability q = 1 − p, the station waits for the beginning of the next time slot and checks the line again.

a. If the line is idle, it goes to step 1.
b. If the line is busy, it acts as though a collision has occurred and uses the backoff procedure.

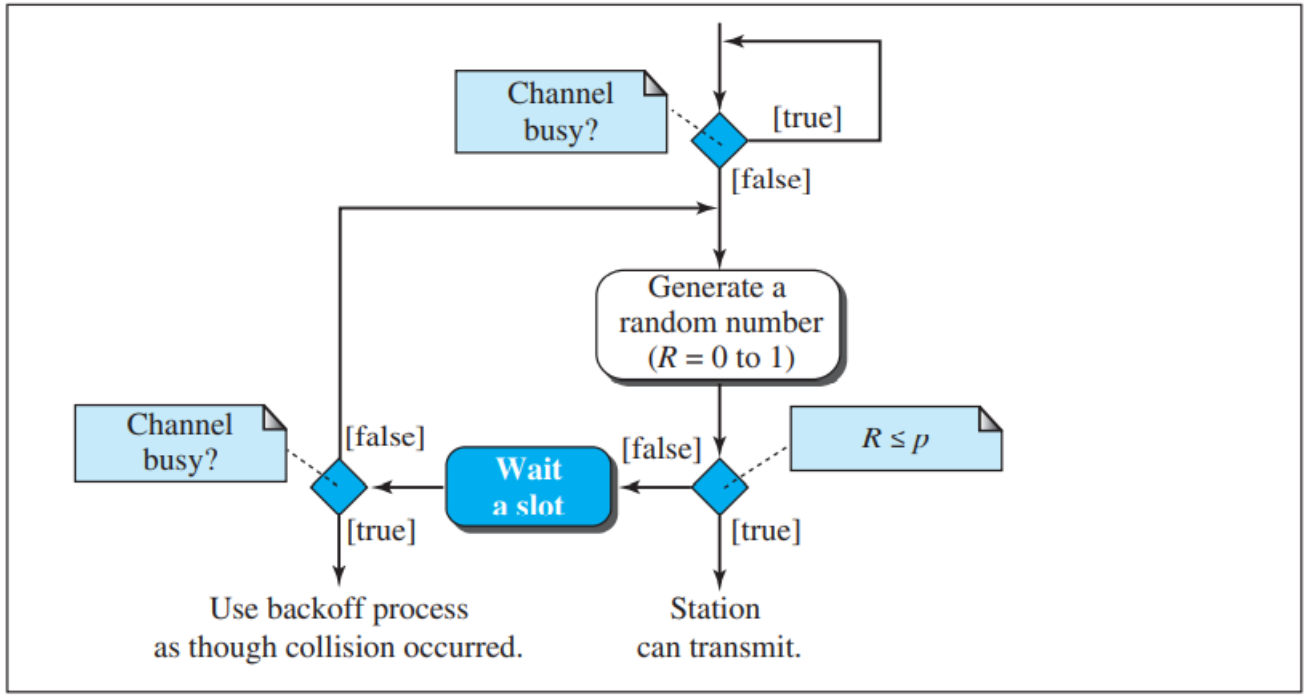
# Flow diagram for three persistence methods




a. 1-Persistent




b. Nonpersistent




c. *p*-Persistent

# CSMA/CD

Carrier sense multiple access with collision detection (CSMA/CD)

# CSMA/CD

- Carrier sense multiple access with collision detection (CSMA/CD) augments the algorithm to handle the collision.
- A station monitors the medium after it sends a frame to see if the transmission was successful. If so, the station is finished.
- If there is a collision, the frame is sent again.

# CSMA / CD

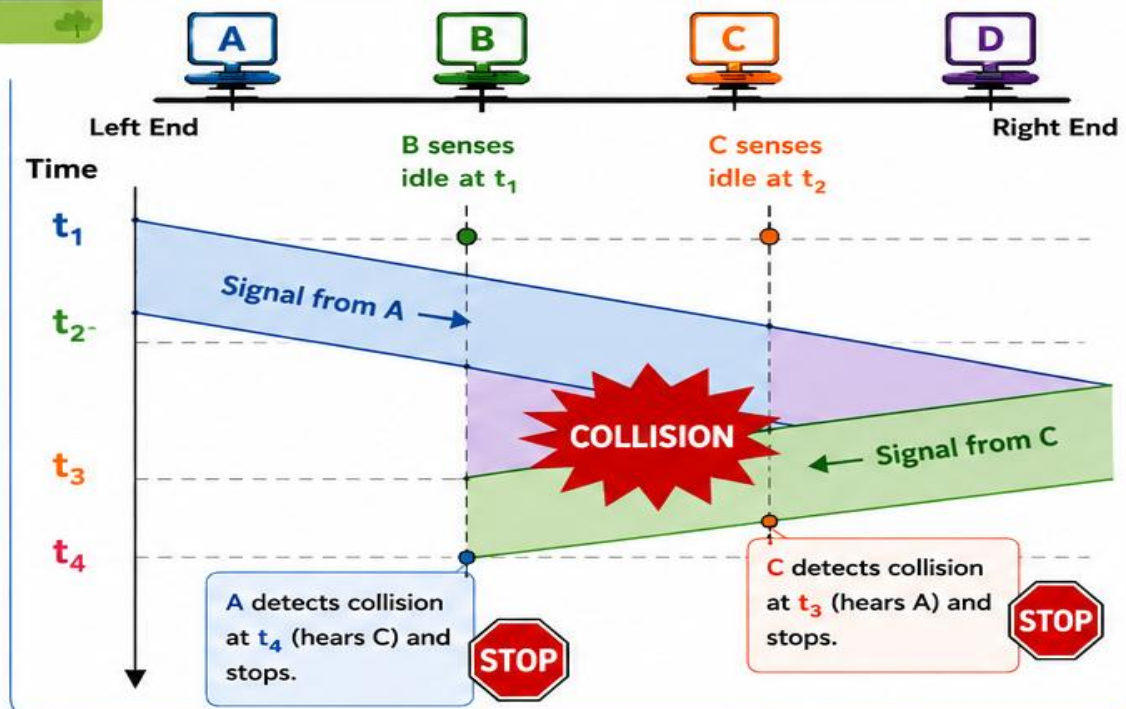
**Listen before talk, Detect collision and Retry**

## REAL-LIFE EXAMPLE




- Cars listen before entering the road.
- If two cars enter at the same time, they collide.
- If a car sees the crash, it stops (detects collision).
- Cars wait for some time and try again.

## WHAT HAPPENS IN THE NETWORK




## WHAT HAPPENS NEXT? (BACKOFF AND RETRY)




### Propagation Delay

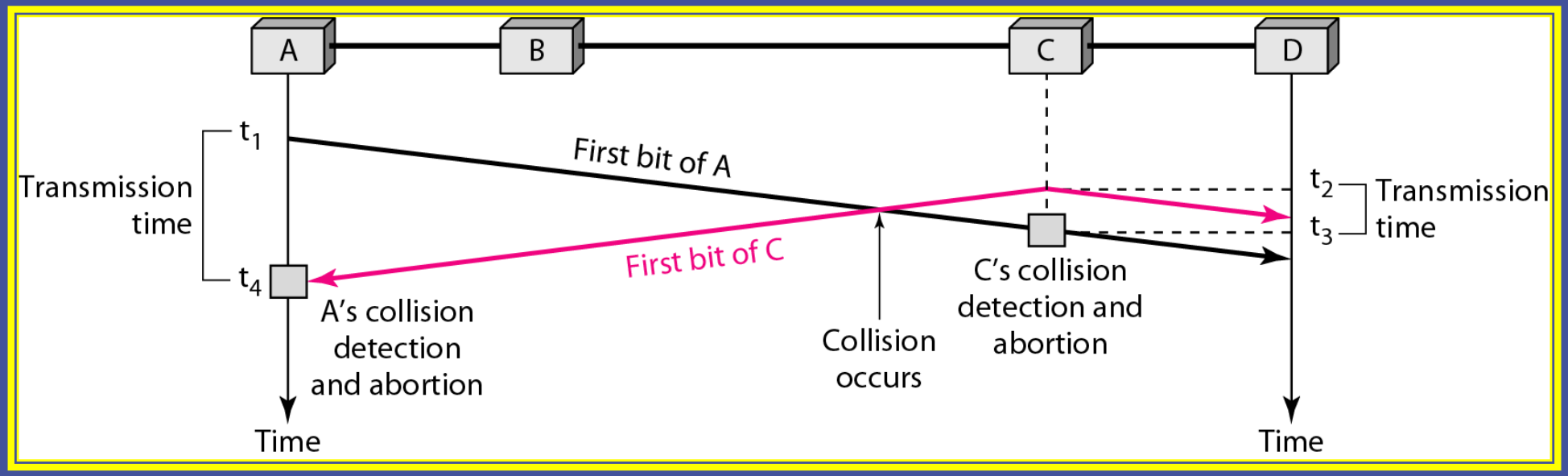
It takes time for a signal to travel from one end of the network to the other.

## STEP-BY-STEP FLOW

1. At time $t_1$, Station A senses the channel idle and starts transmitting.
2. At time $t_2$ ($t_2 > t_1$), Station C also senses the channel idle (A's signal has not reached C yet) and starts transmitting.
3. The two signals meet in the middle of the network and collide.
4. Station C detects the collision at time $t_3$ (when it hears A's signal) and stops transmitting.
5. Station A detects the collision at time $t_4$ (when it hears C's signal) and stops transmitting.
6. Both A and C wait for a random time (backoff) and try again.

**KEY IDEA:** CSMA/CD allows stations to share a channel. It reduces collisions by listening and detecting them, but collisions can still happen due to propagation delay. The system recovers by stopping and retrying.

# Collision of the first bit in CSMA/CD



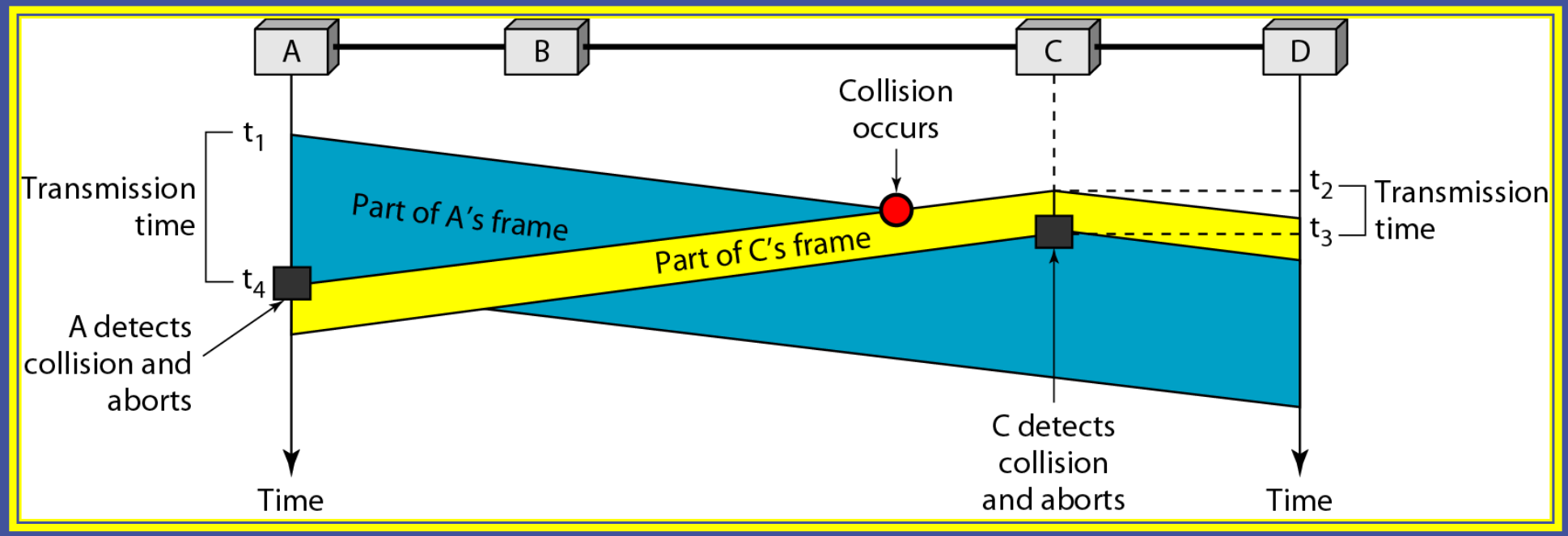
# CONTD...

- Although each station continues to send bits in the frame until it detects the collision, we show what happens as the first bits collide.
- In Figure, stations A and C are involved in the collision.
- At time $t_1$, station A has executed its persistence procedure and starts sending the bits of its frame.
- At time $t_2$, station C has not yet sensed the first bit sent by A.

# CONTD...

- Station C executes its persistence procedure and starts sending the bits in its frame, which propagate both to the left and to the right.
- The collision occurs sometime after time $t_2$.
- Station C detects a collision at time $t_3$ when it receives the first bit of A's frame.
- Station C immediately aborts transmission.
- Station A detects collision at time $t_4$ when it receives the first bit of C's frame; it also immediately aborts transmission.

# Collision and abortion in CSMA/CD



# Contd...

- Minimum Frame Size for CSMA/CD to work, we need a restriction on the frame size.
- Before sending the last bit of the frame, the sending station must detect a collision, if any, and abort the transmission.
- This is so because the station, the frame and does not monitor the line for collision detection.
- The frame transmission time $T_{fr}$ must be at least two times the maximum propagation time $T_p$.

# Contd...

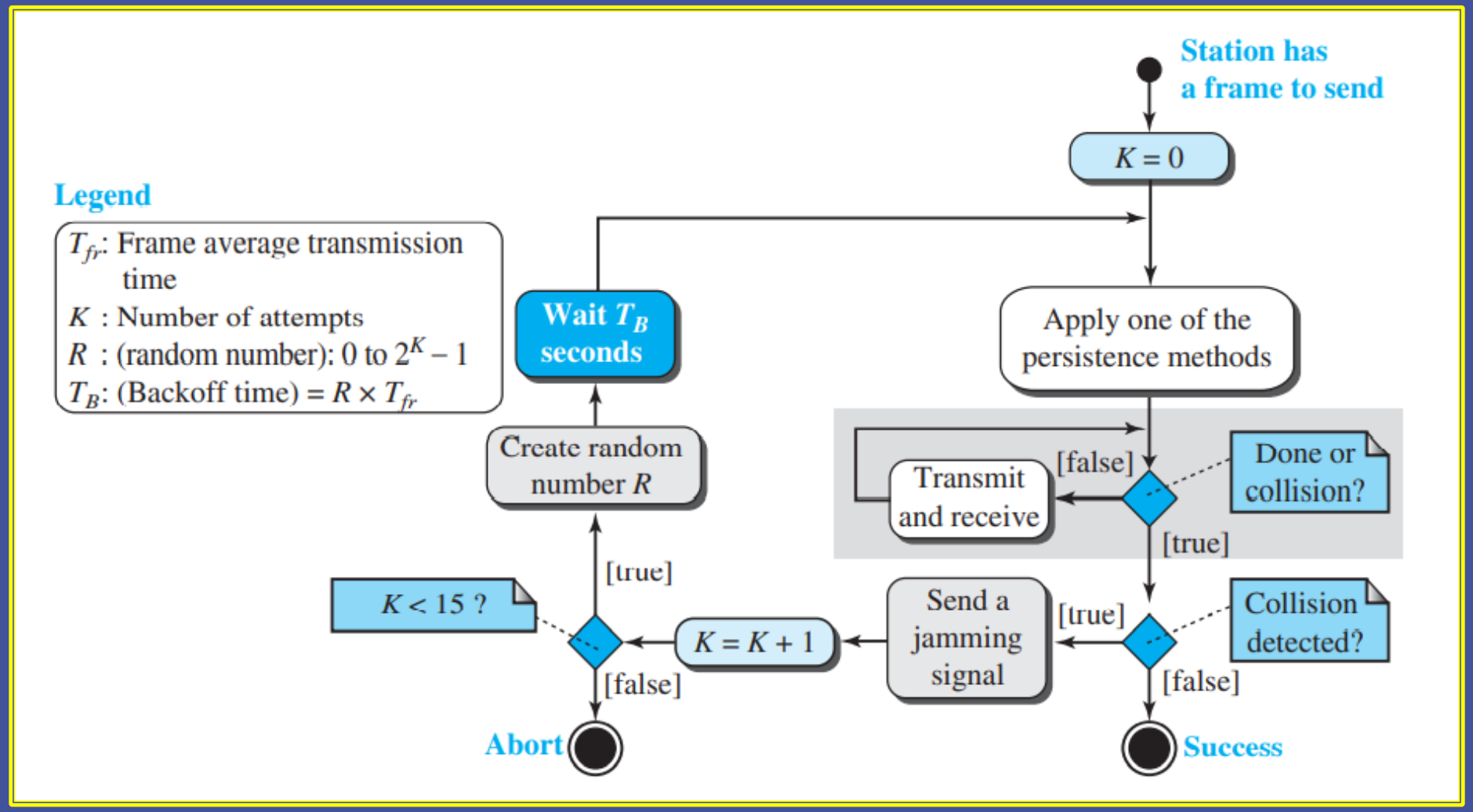
- If the two stations involved in a collision are the maximum distance apart, the signal from the first takes time $T_p$ to reach the second, and the effect of the collision takes another time $T_p$ to reach the first.
- So the requirement is that the first station must still be transmitting after $2T_p$.

# CONTD...

Procedure

- The flow diagram for CSMA/CD in Figure.
- It is similar to the one for the ALOHA protocol, but there are differences.

# Flow diagram for the CSMA/CD



# CONTD...

- The first difference is the addition of the persistence process.
- We need to sense the channel before we start sending the frame by using one of the persistence processes (nonpersistent, 1-persistent, or p-persistent).
- The second difference is the frame transmission. In ALOHA, we first transmit the entire frame and then wait for an acknowledgment.
- In CSMA/CD, transmission and collision detection are continuous processes.

# CONTD...

- We do not send the entire frame and then look for a collision.
- The station transmits and receives continuously and simultaneously (using two different ports or a bidirectional port).
- We use a loop to show that transmission is a continuous process.
- We constantly monitor in order to detect one of two conditions: either transmission is finished or a collision is detected.

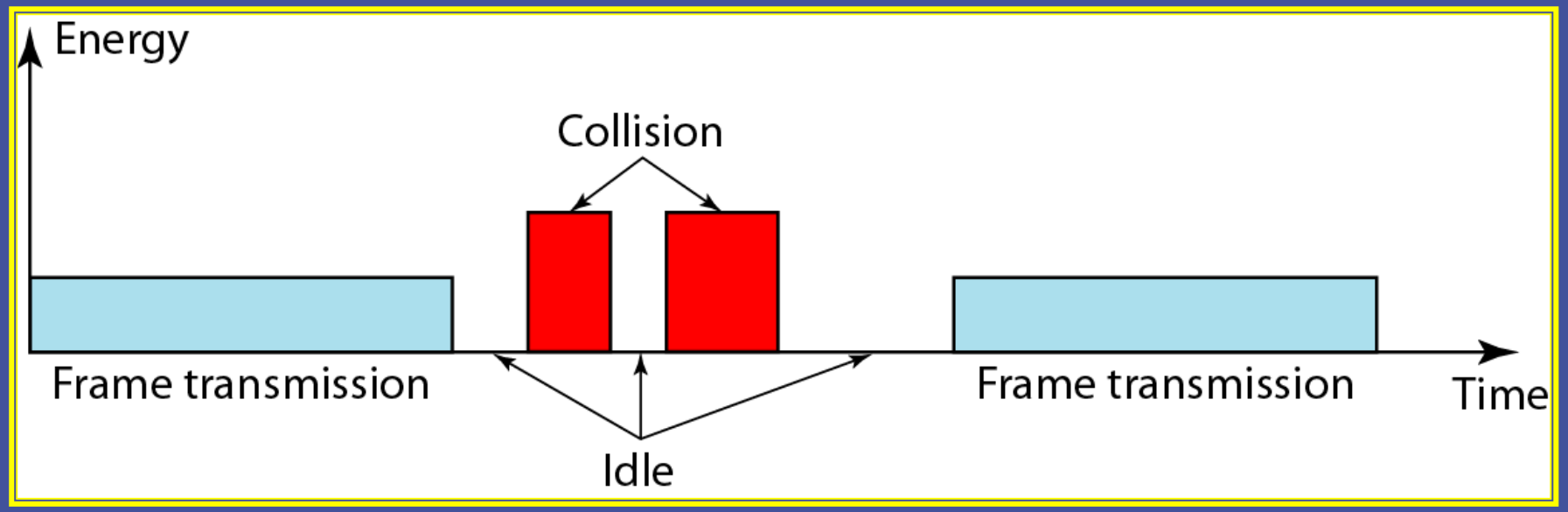
# CONTD...

- Either event stops transmission.
- When we come out of the loop, if a collision has not been detected, it means that transmission is complete; the entire frame is transmitted. Otherwise, a collision has occurred.
- The third difference is the sending of a short jamming signal to make sure that all other stations become aware of the collision.

# Energy Level

- The level of energy in a channel can have three values: zero, normal, and abnormal.
- At the zero level, the channel is idle.
- At the normal level, a station has successfully captured the channel and is sending its frame.
- At the abnormal level, there is a collision and the level of the energy is twice the normal level.
- A station that has a frame to send or is sending a frame needs to monitor the energy level to determine if the channel is idle, busy, or in collision mode.

# Energy level during transmission, idleness, or collision



# Throughput

- The throughput of CSMA/CD is greater than that of pure or slotted ALOHA.
- The maximum throughput occurs at a different value of G and is based on the persistence method and the value of p in the p-persistent approach.
- For the 1-persistent method, the maximum throughput is around 50 percent when G = 1.
- For the nonpersistent method, the maximum throughput can go up to 90 percent when G is between 3 and 8.

# Traditional Ethernet

- One of the LAN protocols that used CSMA/CD is the traditional Ethernet with the data rate of 10 Mbps.

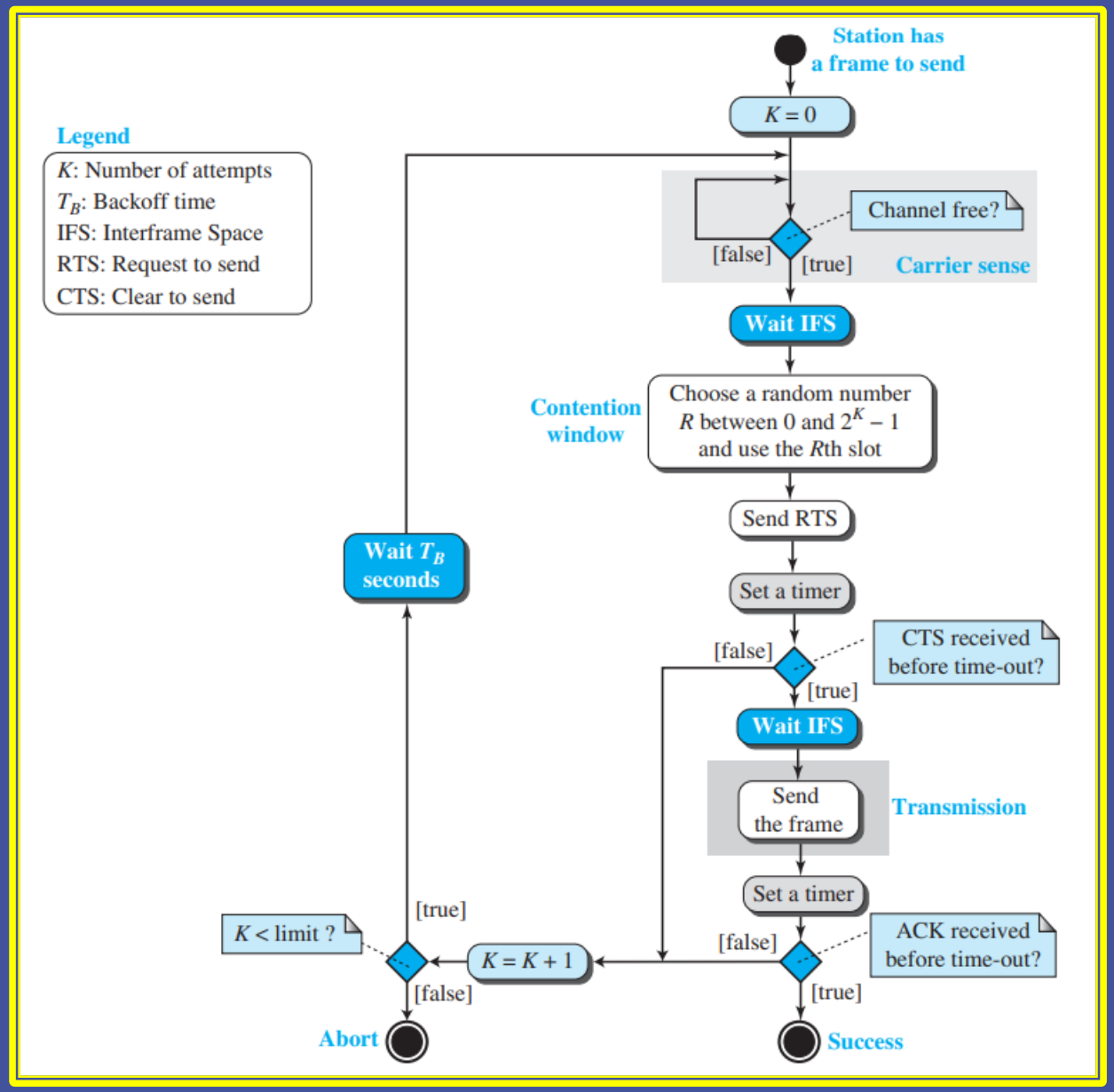
# CSMA/CA

Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)

# CSMA/CA

- Carrier sense multiple access with collision avoidance (CSMA/CA) was invented for wireless networks.
- Collisions are avoided through the use of CSMA/CA's three strategies: the interframe space, the contention window, and acknowledgments.

# Flow diagram of CSMA/CA



Interframe Space (IFS)

Contention Window

Frame Exchange Time Line



Network Allocation Vector

Collision During Handshaking

Hidden-Station Problem

# Interframe Space (IFS)

- First, collisions are avoided by deferring transmission even if the channel is found idle.
- When an idle channel is found, the station does not send immediately.
- It waits for a period of time called the interframe space or IFS.
- The channel is idle when it is sensed, a distant station may started transmitting.

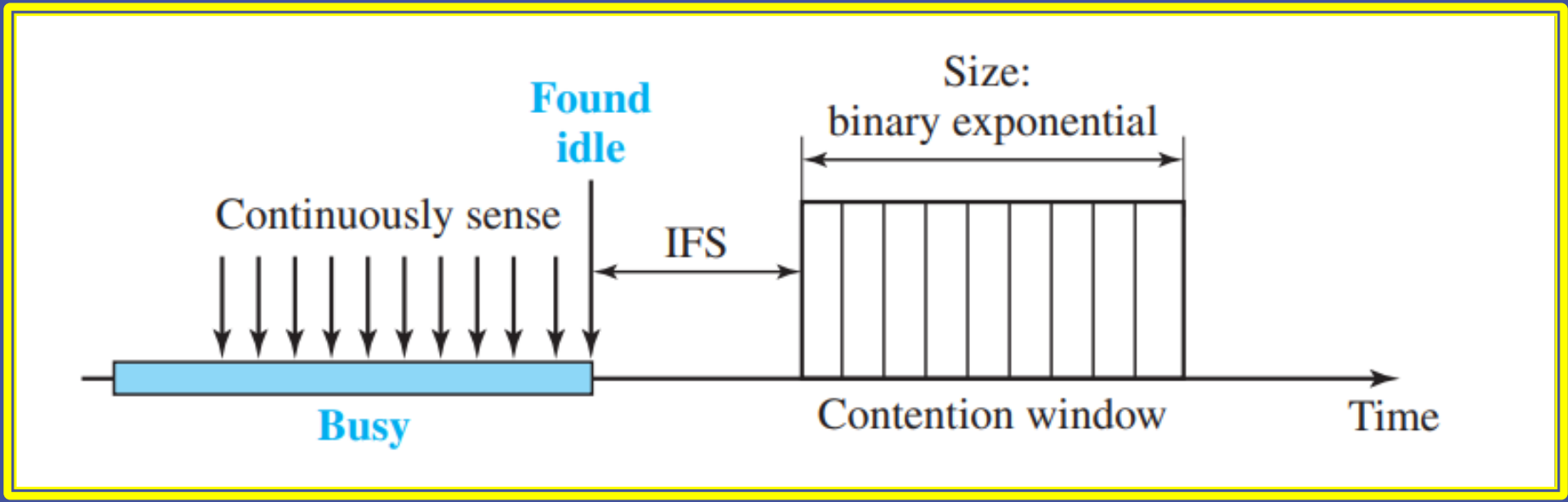
# Contd…

- The distant station's signal has not yet reached this station.
- The IFS time allows the front of the transmitted signal by the distant station to reach this station.
- After waiting an IFS time, if the channel is still idle, the station can send, but it still needs to wait a time equal to the contention window.
- For example, a station that is assigned a shorter IFS has a higher priority.

# Contention Window

- The contention window is an amount of time divided into slots.
- A station that is ready to send chooses a random number of slots as its wait time.
- The number of slots in the window changes according to the binary exponential backoff strategy.
- This means that it is set to one slot the first time and then doubles each time the station cannot detect an idle channel after the IFS time.

# Contention window



# Contd…

- This is very similar to the p-persistent method except that a random outcome defines the number of slots taken by the waiting station.
- In contention window, the station needs to sense the channel after each time slot.
- If the station finds the channel busy, it does not restart the process; it just stops the timer and restarts it when the channel is sensed as idle.
- This gives priority to the station with the longest waiting time.

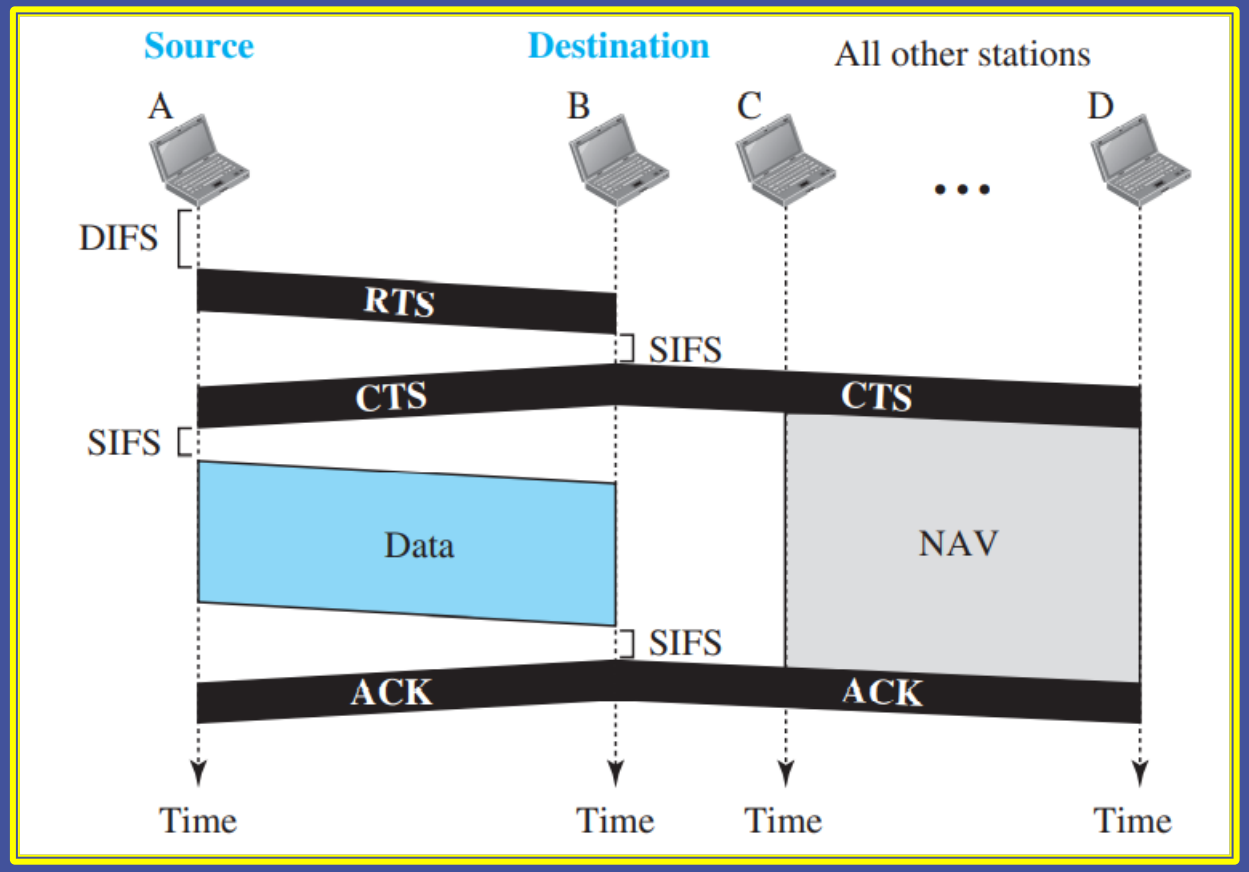
# Frame Exchange Time Line – 4 Steps

- Before sending a frame, the source station senses the medium by checking the energy level at the carrier frequency.
  - The channel uses a persistence strategy with backoff until the channel is idle.
  - After the station is found to be idle, the station waits for a period of time called the DCF interframe space (DIFS); then the station sends a control frame called the request to send (RTS).

# Contd…

- After receiving the RTS and waiting a period of time called the short interframe space (SIFS), the destination station sends a control frame, called the clear to send (CTS), to the source station.
- This control frame indicates that the destination station is ready to receive data.

# CSMA/CA and NAV



# Contd…

- The source station sends data after waiting an amount of time equal to SIFS.
- The destination station, after waiting an amount of time equal to SIFS, sends an acknowledgment.
- Acknowledgment is needed in this protocol because the station does not have any means to check for the successful arrival of its data at the destination.
- The lack of collision in CSMA/CD is a kind of indication to the source that data have arrived.

# Network Allocation Vector

- The key is a feature called NAV. When a station sends an RTS frame, it includes the duration of time that it needs to occupy the channel.
- The stations that are affected by this transmission create a timer called a network allocation vector (NAV) that shows how much time must pass before these stations are allowed to check the channel for idleness.

# Contd…

- Each time a station accesses the system and sends an RTS frame, other stations start their NAV.
- Each station, before sensing the physical medium to see if it is idle, first checks its NAV to see if it has expired.

# Collision During Handshaking

- If there is a collision during the time when RTS or CTS control frames are in transition, often called the handshaking period?
- Two or more stations may try to send RTS frames at the same time.
- These control frames may collide.
- There is no mechanism for collision detection, the sender assumes there has been a collision if it has not received a CTS frame from the receiver.
- The backoff strategy is employed, and the sender tries again

# Hidden-Station Problem

- The solution to the hidden station problem is the use of the handshake frames (RTS and CTS).
- Because both B and C are within the range of A, the CTS message, which contains the duration of data transmission from B to A, reaches C.
- Station C knows that some hidden station is using the channel and refrains from transmitting until that duration is over.

# Conclusion

- CSMA/CA was mostly intended for use in wireless networks.
- Support to handle issues related to wireless networks, such as hidden terminals or exposed terminals